

Ana Aquiles [Website](#)

Data Scientist · Computational Neuroscience · ML/DL · Python



Ana Aquiles

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MEXICAN, 30-AUGUST-1995

PROFESSIONAL SUMMARY

I am a data scientist with a PhD specialized in Computational Neuroscience and **8+ years of experience** converting large, high-dimensional time-series and imaging datasets into actionable insights. I specialize in building end-to-end Python pipelines using machine learning, deep learning, and statistical methods. My experience spans various data domains, including neural signals, movie recordings, MRI volumes, and stochastic time series, allowing me to apply a *domain-agnostic approach with techniques such as dimensional reduction and anomaly and pattern detection*. I am skilled at managing projects from data design through to modeling and effectively communicating findings to non-technical stakeholders. **I am eager to apply my quantitative skills in a Data Scientist role beyond of academia .**

CORE TECHNICAL SKILLS

Python (Advanced)	Machine Learning	Deep Learning (TF/PyTorch)	Time-Series Analysis
Statistical Modelling	Data Visualization	Signal Processing	Dimensionality Reduccion
ETL/ Data Pipelines	Large-scale Datasets	Image Analysis (openCV)	Multivariate models
LLM-assisted API	Reproducible/Open Science	Unix-like operating systems	Team Mentoring

Libraries & Tools: Numpy . SciPy . Pandas. Scikit-learn. TensorFlow/PyTorch. Matplotlib. Seaborn. OpenCV. NiBabel (MRI). SQL . Git/Github. Claude API

Applied Methods: Population/State-Space Geometry . Information Theory . Geometric/ Topological Data Analysis . Classification & Clustering . Multivariate time-series forecasting. LLM-Assisted Code & Analysis Development

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher - Computational Neuroscience

Apr 2024 - Mar 2026

CNRS, Institut du Génomique Fonctionnelle, Montpellier, France.

- Built ML and deep-learning pipelines in Python to process large-scale, high-frequency time-series and video datasets (>10 GB per experiment), engineering the full path from raw signal to model-ready features.
- Applied dimensionality-reduction and population/state-space geometry methods to uncover directed coupling and bistable dynamics in multivariate time-series — a technique directly transferable to anomaly detection and regime-change forecasting in any complex sequential dataset (published, bioRxiv 2026).
- Integrated Claude (Anthropic LLM API) into the research workflow to accelerate deep-learning code development and to design a null-generative model producing synthetic data for statistical significance testing, strengthening the rigor of model validation.
- Mentored team members on Python and ML best practices; led internal workshops on applied machine-learning methods.

Research Data Engineer - Computational Neuroscience

Sep 2023 - Apr 2024

CNRS, Institut du Génomique Fonctionnelle, Montpellier, France.

- Designed and implemented data pipelines for high-throughput, multi-channel imaging data, including quality-control and batch-processing scripts to guarantee integrity at scale.
- Partnered across institutions to deliver standardised, reproducible data workflows adopted by multiple research teams.

PhD Researcher - Computational & Systems Data Analyst
UNAM, Neurobiology Institute. Brain Connectivity Laboratory, Queretaro, Mexico.

Aug 2019 - Dec 2023

- Designed and built a full analytical pipeline integrating three distinct time-series/imaging modalities (electrophysiology, local field potentials, MRI) into a single comparable feature space — published in Scientific Reports (2023) and iScience (2025).
- Applied population-dynamics, machine-learning, and information-theory methods to high-dimensional electrophysiological time-series as part of a cross-institutional collaboration (visiting researcher, 2023).
- Supervised and trained 4 MSc/BSc students on data-analysis projects; co-authored 5 peer-reviewed publications.

Research Assistant - Computational & Systems Data Analyst
UNAM, Neurobiology Institute. Brain Connectivity Laboratory, Queretaro, Mexico.

Jan 2018 - Aug 2019

- Led data acquisition and analysis for imaging/diffusion-tensor studies; built automated processing scripts in Python.
- Trained postgraduate students in image-analysis workflows and experimental protocol design.

EDUCATION

PhD in Biomedical Sciences
Neurobiology Institute, UNAM CAMPUS, Juriquilla. Queretaro | 08/2019 - 12/2023
Bachelor of Science in Biology
Faculty of Sciences, UNAM, Mexico | 08/2014 - 01/2018

KEY TRAINING AND CERTIFICATIONS

L'apprentissage automatique (machine learning) appliqué à l'analyse- BIOCAMPUS
Postdoctoral Scientist

Jun 2025

ib Cegos. Python Perfectionnement- CNRS Formation
Postdoctoral Scientist

May 2025

NEUBIAS workshop in bioimage analysis at Pasteur Institute, France
Postdoctoral Scientist

May 2025

Computational methods and application into segmentation and image treatment using Python

Neuromatch Computational Neuroscience school, online
Ph.D student.

Jul 2020

LASCON (Latin American School on Computational Neuroscience), Brazil

Jan 2020

Ph.D student.
Computational neuroscience methods and application to real data and models.

UV, University of Valparaiso. Fundacion Ciencia y Vida. Chile
BS. in Biology, Internship
Data Analysis and neurocomputation at Computational Neuroscience Lab lead by Dr. Patricio Orio

Dec 2016- Feb 2017

LANGUAGES

Spanish - native **English**- proficient **French**- proficient **Portugues**- intermediate